The large majority of the questions worked well in providing the required success rate for the paper and a satisfactory discrimination between the candidates. Only one question, namely **Question 20**, proved to be very easy.

The questions which caused most difficulty were **Questions 13, 14, 27** and **36**.

Comments on Individual Questions

Question 13

This question's four alternatives were equally popular suggesting that a large proportion of the candidates were guessing. In comparing reaction 2 with reaction 1 two factors had to be considered; particle size and the mass of magnesium. The former factor produced an increase in rate and the latter produced only half the volume of hydrogen. Consequently the graphs in alternative D correctly represented the two reactions.

Question 14

The alternatives C and D were each chosen by approximately forty percent of the entry. Among the ideas being tested in this question were firstly that a catalyst does take part in a reaction and is unchanged at the end of a reaction. Secondly that a catalyst only speeds up a reaction but does not start the reaction.

Question 17

The responses to this question were very varied. The key to success in this question was the realisation that calcium sulfate is insoluble in water and would be prepared by a method involving precipitation. In general, precipitation reactions involve the mixing together of two solutions each of which contain one of the ions (the ions being different) in the insoluble product.

Question 22

From the proton numbers given it was possible, using the Periodic Table, to identify each of the four elements given in the question. Only one of the elements was a transition metal and a typical property of transition metals and their compounds is their ability to act as catalysts. Thus alternative C was the correct answer.

Question 26

Copper is below hydrogen in the reactivity series and therefore does not react with dilute hydrochloric acid. Thus alternative D which had a red-brown residue, copper, was the answer to the question.

Question 30

This was a recall question, straight from the syllabus. Although many correct answers were seen, a quarter of the entry incorrectly opted for alternative A. Candidates should remember that not every multiple choice question requires powers of deduction.

A large number of questions had high success rates but nevertheless, the paper overall provided a satisfactory test of each candidate's ability. **Question 34** did not discriminate well between the candidates. Apart from **Question 34** all the questions performed well and showed good discrimination.

Comments on Individual Questions

Question 1

This question was solved by realising that hydrogen, in the list of gases, was the only gas less dense than air. Therefore the only gas which could be collected by the method shown in the question.

Question 16

Almost half the entry incorrectly suggested that a catalyst does not take part in a reaction. Catalysts do take part in reactions but are unchanged at the end of the reactions.

Question 17

As a general rule an insoluble salt e.g. calcium sulfate is formed by a precipitation reaction which involves the mixing together of solutions of two soluble salts. Only alternative B contained two soluble salts with all the other alternatives containing at least one insoluble salt.

Question 18

Titration as a method of forming salts can only be used for the preparation of soluble salts, since it is very difficult in a reaction involving the formation of an insoluble salt to determine when precipitation is complete.

Question 21

Sodium floats on water and therefore must have a density less than 1g/cm^3. The density values for the elements provided the quick method for identifying sodium.

Question 25 and Question 27

Unfortunately, to some extent these two questions overlapped and in both a failure to realise that copper is below hydrogen in the reactivity series was the downfall of many candidates. Due to its position in the reactivity series, copper does not react with either dilute hydrochloric or dilute sulfuric acid.

Question 33

Calcium carbonate undergoes decomposition on heating to form carbon dioxide and consequently many of the candidates mistakenly included calcium carbonate in their answer. The word burn was the key word in the question and calcium carbonate does not burn.

Question 36

Careful examination of alternative A reveals that it is not a carboxylic acid. However a significant number of candidates did fail to notice that the groups shown were not carboxyl groups.

The Alternative to Practical Chemistry paper is designed to test the candidate's knowledge and experience of practical chemistry.

Skills tested include recognition, use and calibration of chemical apparatus, recall of experimental procedures, handling and interpretation of data, drawing of graphs, analysis of unknown salts and calculations.

The standard continues to be maintained and the majority of candidates show evidence of possessing many of the aforementioned skills.

Most candidates show competency of plotting points accurately on graphs and joining the points as instructed.

Calculations are generally completed accurately using the appropriate significant figures.

Comments on specific questions

Question 1

(a) Aqueous ammonia is an alkaline solution and therefore turns litmus solution blue.

(b) Aqueous copper(II) sulfate is a blue solution.

(c) When aqueous ammonia is added to aqueous copper(II) sulfate a blue precipitate is formed which dissolves in excess ammonia forming a deep blue solution.

Question 2

(a) The pink solid is copper which is deposited at electrode **B**. This is because copper ions are positively charged and therefore move to the negative electrode.

(b) The gas formed is hydrogen. When testing for hydrogen a burning (not glowing) splint is used and a pop is heard.

(c) The gas formed is oxygen which relights a glowing splint.

In **(b)** and **(c)** credit was not given if the gases were given at the wrong electrodes.

As sulfate ions were present in the solution a correct test for sulfur dioxide was also given credit.

(d) During the electrolysis the colour changes from blue to colourless as the copper ions are removed from the solution.

(a) The mass of iron(II) sulfate crystals used in the experiment is 3.85 g.

(b) (i) When the crystals lose their water of crystallisation they are described as anhydrous. Dehydrated was also accepted.

(ii) The mass of iron(II) sulfate which remained after heating is 2.30 g.

(iii) The mass of water lost from the crystals is 1.55 g.

Errors in **(a)** and **(b)** can be carried forward and if used correctly this mark can be gained. This applies throughout the paper to all questions where previous incorrect answers are used.

(c) The relative formula masses of iron(II) sulfate and water are 152 and 18. Both answers are required for the mark.

(d) 0.015 moles of iron(II) sulfate remained after heating.
0.086 moles of water were lost during heating.

Candidates who gave their answers to only one significant figure did not receive credit.

(e) The value of **x** is 5.7 and the formula is $FeSO_4.6H_2O$.

To score the first mark 5.7 had to appear on the paper although it could then be rounded up to 6 on the answer line.

Questions 4 to 6

Correct answers to the multiple choice questions are **(c)**, **(b)**, **(c)**.

Question 7

There are two scoring points. The graph should be steeper than the given line because the acid is more concentrated and the horizontal line should be at twice the volume of gas than the given line because there are twice as many moles of acid present. Most candidates scored the first point only.

Question 8

(a) The colour change of phenolphthalein was from pink to colourless. These colours were given in the question and no others were accepted.

(b) The three titres are 26.5 (or 26.6), 25.8 and 26.0. As usual when errors occur in reading the burette diagrams or subtracting the volumes the mean must be taken from the closest two titres. A common error was to use all three titres in calculating the mean.

Answers to calculations are:

(c) 0.0025

(d) 0.0025

(e) 0.0965 (0.096 and 0.097 were both accepted)

(f) 88

As before, errors may be carried forward. This also applies to **(g)** and **(h)**.

(g) The value of n is 3. It is obtained by subtracting 46 from the answer to **(f)** and dividing by 14. This makes the formula of the acid $C_3H_7CO_2H$.

Question 9

This question involves the analysis of zinc chloride.

(a) A colourless solution is formed as transition metal ions are not present.

(b) Solutions containing aluminium or zinc ions produce a white precipitate when aqueous sodium hydroxide is added. The precipitate is soluble in excess sodium hydroxide.

(c) When aqueous ammonia is added to a solution containing zinc ions a white precipitate is formed. This precipitate is soluble in excess aqueous ammonia.

(d) Chloride ions are confirmed by adding nitric acid and aqueous silver nitrate. A white precipitate is formed. No marks could be obtained if silver nitrate was not used. The use of acidified silver nitrate lost 1 mark but if hydrochloric acid was used only the white precipitate mark was given.

Question 10

(a) Candidates are asked to read the thermometers and complete the table. The temperatures are 27, 31 and 32.

(b) The results are plotted on the graph. Candidates are given credit for plotting the points accurately and joining them with two intersecting straight lines. Most candidates scored well.

(c) The answers are read from the candidate's graph. If the graph is correctly drawn the answers are 33.8 and 22.

(d) The concentration of the sulfuric acid was 0.45 mol/dm^3. This is based on a correct mole ratio in **(i)**. Marks could still be obtained for a calculation based on an incorrect equation.

(e) Most candidates gained credit for stating that the reaction had finished but failed to state that the temperature decreased due to the addition of the cooler sulfuric acid.